Daniel Waxman

Basis Research Institute & MIT
Cambridge, MA

dan@basis.ai / dwaxman@mit.edu
danwaxman.github.io

EDUCATION

Ph.D. Electrical Engineering, *December 2025*
Stony Brook University
Thesis: *Sequential Bayesian Learning and Ensembles for Online Inference and Experimental Design*
Advisor: Petar M. Djurić

B.S. Mathematics & Applied Mathematics and Statistics *(cum laude)*, May 2021
Stony Brook University

EMPLOYMENT & ACADEMIC APPOINTMENTS

01/26 – Present Postdoctoral Research Scientist
Basis Research Institute, Cambridge, MA.
Jointly-supervised postdoctoral fellowship with by Matthew Levine (Basis) and Youssef Marzouk (MIT).

05/26 – Present Research Affiliate
Laboratory for Information & Decision Systems
Massachusetts Institute of Technology, Cambridge, MA.
Member of the MIT Unertainty Quantification Group as part of jointly-supervised postdoctoral fellowship.

06/25 – 01/26 Research Intern
Basis Research Institute, New York, NY.
Research in causal inference, probabilistic programming, and applied Bayesian machine learning with Rafal Urbaniak and Jack Feser.

RESEARCH INTERESTS

I'm interested in Bayesian statistics and machine learning, broadly construed. This has most often taken the form of Gaussian processes, dynamical systems, and ensembling methods. I've also worked on causal inference, including applications to dynamical systems and mechanistic interpretability.
As of late, I've been particularly interested in PPLs & Bayesian workflow for dynamical systems, and developing a post-Bayesian & predictively-oriented theory of inference for dynamical systems.

PUBLICATIONS

* denotes equal contribution, † denotes a student or intern primarily supervised by me

Preprints and Articles In Review

2026 R. Urbaniak, S. Witty, **D. Waxman**, A. Zane, P. Garg, E. Bunnapradist, S. Vaidyanathan, J. Feser, D Lehe, E. Bingham, "A Computationally Feasible Framework for Causal Probabilistic Explanation."
[arXiv] [code]

2026 L. Snow†, **D. Waxman**, M. Levine, "Wiener-Space Cubature for Continuous-Discrete Filtering." *Submitted.*
[arXiv] [code]

2026 F. Sergeev, M. Heinonen, **D. Waxman**, T. Cooijmans, R. Baptista, D. Batenkov, E. Bingham, "Simulation-Free Learning of Population Dynamics with Wasserstein Lagrangian Residuals." *Submitted.*
[arXiv] [code]

2026 S. Vaidyanathan*, R. Urbaniak*, E. Bunnapradist, M. Naim, and **D. Waxman**. "Slaying the Hydra: Uncovering Backup Mechanisms in Language Models." *Submitted.*
[arXiv]

2026 **D. Waxman**, D. Batenkov, J. Feser, A. Zane, E. Bingham, Y. Marzouk, M. Levine, "Dynestyx: A Probabilistic Programming Library for Dynamical Systems." *Submitted.*
[arXiv] [code]

Journal Papers

2026 M. Levine, **D. Waxman**, I. Urteaga. "CD-Dynamax: A JAX-based Python package for continuous-discrete probabilistic state space modeling and inference." *Journal of Open Source Software*.
[pdf] [code]

2026 K. Lamer, E. V. Davis, Z. Mages, Z. Zhu, N. E. Rojas-Robles, K. N. Mitu, E. R. Vivoni, Z. Wang, H. Li, M. Cadeddu, P. Joshi, **D. Waxman**. "The Encanto Neighborhood Field Experiments Dataset: Land–Atmosphere Interactions in Urban Phoenix." *Scientific Data*, 2026.
[SpringerNature]

2026 **D. Waxman**, F. Llorente, P. M. Djurić, "Sequential Inference with Gaussian Processes: A Signal Processing Perspective." *IEEE Signal Processing Magazine*, 2026.
[IEEE Xplore] [arXiv]

2026 **D. Waxman**, F. Llorente, P. M. Djurić. "Bayesian Ensembling: Insights from Online Optimization and Empirical Bayes." *Transactions on Machine Learning Research (TMLR)*, 2026.
[OpenReview] [arXiv] [code]

2025 **D. Waxman**, F. Llorente, K. Lamer, P. M. Djurić. "Designing an Optimal Sensor Network via Minimizing Information Loss." *Bayesian Analysis, 2025.*
[ProjectEuclid] [arXiv] [code]

2024 **D. Waxman**, P. M. Djurić. "Dynamic Online Ensembles of Basis Expansions." *Transactions on Machine Learning Research (TMLR)*, 2024.
[OpenReview] [arXiv] [code]

2024 **D. Waxman**, K. Butler, P. M. Djurić. "Dagma-DCE: Interpretable, Non-Parametric Differentiable Causal Discovery." *IEEE Open Journal of Signal Processing*, vol. 5, pp. 393-401, 2024.
[IEEE Xplore] [arXiv] [code]

Conference Papers

2026 **D. Waxman**, M. Levine, K. Butler, P.M. Djurić, E. Mackevicius, D. Batenkov. "Bayesian Learning of Graph-Based Interaction Potentials with Applications to Studying Animal Behavior." To appear in *Proceedings of the 2026 Asilomar Conference on Signals, Systems, and Computers, 2026*.

2026 M. Heinonen, Y. Shenfeld, R. Baptista, **D. Waxman**, D. Batenkov, T. Cooijmans, E. Bingham. "Wasserstein Residuals: Learning Gradient Flows from Population Dynamics." In *Advances in Neural Information Processing Systems, 39, 2026*
[OpenReview] [arXiv] [code]

2026 F. Llorente, **D. Waxman**, S. Jantre, N. Urban, S. Minkoff. "Robust, Online, and Adaptive Decentralized Gaussian Processes." In *Proceedings of the 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
[IEEEXplore] [arXiv]

2025 **D. Waxman**, P. M. Djurić. "Non-Stationary Causal Learning via Hierarchical Modeling." In *Proceedings of the 2025 Asilomar Conference on Signals, Systems, and Computers, 2025.*
[IEEEXplore]

2025 F. Llorente*, **D. Waxman***, P. M. Djurić. "Decentralized Online Ensembles of Gaussian Processes for Multi-Agent Systems." In *Proceedings of the 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
[IEEEXplore] [arXiv] [code]

2024 K. Butler*, **D. Waxman***, P. M. Djurić. "Tangent Space Causal Inference: Leveraging Vector Fields for Causal Discovery in Dynamical Systems." In *Advances in Neural Information Processing Systems, 37, 120078-120102, 2024.*
[OpenReview] [arXiv] [code]

2024 **D. Waxman**, P. M. Djurić. "A Gaussian Process-based Streaming Algorithm for Prediction of Time Series With Regimes and Outliers." In *Proceedings of the 27th International Conference on Information Fusion (FUSION), Venice, Italy, 2024*.
[IEEEXplore] [arXiv] [code]

2023 M. Ajirak, **D. Waxman**, F. Llorente, P. M. Djurić. "Fusion of Gaussian Process Predictions with Monte Carlo Sampling." In *Proceedings of the 57th Asilomar Conference on Signals, Systems, and Computers (pp. 1367-1371), 2023*.
[IEEE Xplore] [arXiv]

2023 Y. Liu, C. Cui, **D. Waxman**, K. Butler, P. M. Djurić. "Detecting Confounders in Multivariate Time Series Using Strength of Causation." In *Proceeedings of the 31st European Signal Processing Conference (EUSIPCO), pp. 1400-1404, 2023*.
[IEEE Xplore] [pdf] [video]

Workshop Proceedings & Professional Meeting Abstracts

2026 A. Arora, R. E. Peterson, E. L. Mackevicius, **D. Waxman**, M. Levine, "Continuous-Discrete Switching Dynamical Model for Multi-Agent Interactions." In *AI for Stochastic Dynamics (STODY) Workshop @ NeurIPS 2026*.
[OpenReview]

2026 K. Lamer, E. V. Davis, Z. J. Mages, P. Joshi, N. Rojas, E. R. Vivoni, Z. Wang, Z. Zhu, **D. Waxman**. "Multisensor Observations and Meter-Scale Modeling of Urban Park Cooling Dynamics for Digital Twin Integration." *106th AMS Annual Meeting*.

2024 D. Chernoff, **D. Waxman**, G. Brite, L. Langman, C Rabeno, L. Czerwonka, P. M. Djurić. "The Development of an Automated Algorithm to Identify and Manage Post-thyroidectomy Hypocalcemia." *American Head and Neck Society 2024 General Meeting*.

RESEARCH TALKS

Invited Talks

2027 BayesComp '27
j-ISBA Sponsored Session
Talk Title: "Towards A Bayesian Workflow for Dynamical Systems"

2027 SIAM Conference on Computational Science and Engineering (CSE) '27
Minisymposium on "Beyond Standard Bayesian Paradigms for Reliable Uncertainty Quantification"
Talk Title: "Towards a Theory of Predictively-Oriented Filtering"

2026 Dartmouth Computational and Applied Mathematics Seminar
Talk Title: "`dynestyx`: A Probabilistic Programming Language for Dynamical Systems"

2026 MIT Physical Mathematics Seminar
Talk Title: "`dynestyx`: A Probabilistic Programming Language for Dynamical Systems" [link]

2026 MIT Computational Science and Engineering Community Seminar
Talk Title: "`dynestyx`: A Probabilistic Programming Language for Dynamical Systems" [link]

2026 SIAM New England Section Annual Meeting '26
Minisymposium on "Generative Scientific Machine Learning for PDEs and Complex Physical Systems"
Talk Title: "A Bayesian Recipe for Learning Hybrid Models from Noisy Data"

2026 ISBA World Meeting '26
Session on "Calibrated Bayes: Model Design and Adaptation Under Limited Resources"
Talk Title: "Online & Predictively-Oriented Model Fusion"

2026 SIAM Uncertainty Quantification '26
Minisymposium on "Sequential Inference for Dynamical Systems"
Talk Title: "GP4SP: Sequential Gaussian Process Inference for Signal Processing"

2026 Queens College Computer Science Colloquium
Talk Title: "Online Bayesian Learning and Ensembles" [slides]

2023 Acoustics Research Institute of the Austrian Academy of Sciences
Talk Title: "Causal Discovery via Quantifying Influences" [link] [slides]

2023 Bellairs Workshop on Machine Learning and Statistical Signal Processing for Data on Graphs
Talk Title: "Bayesian Combination"

Contributed Talks

2024 SIAM New York-New Jersey-Pennsylvania Section Conference '24
Talk Title: "Optimizing Observation Locations via Minimizing Information Loss"

SELECTED HONORS AND AWARDS

2026 Best Poster Award @ Bayesian Young Statisticians Meeting (BAYSM) 2026

2026 Gold Reviewer @ ICML 2026

2025 Top Reviewer @ NeurIPS 2025

TEACHING

As Instructor of Record

Fall 2025 Understanding Machine Learning (ESE 188 @ SBU)

Spr 2025 Understanding Machine Learning (ESE 188 @ SBU)

As Teaching Assistant

Sum 2022 Random Signals and Systems (ESE 306 @ SBU)
Spr 2022 Random Signals and Systems (ESE 306 @ SBU)
Fall 2021 Programming Fundamentals (ESE 124 @ SBU)

OUTREACH & SERVICE

Reviewing

I've served as a reviewer for 60+ manuscripts across top-tier journals and conferences in machine learning and signal processing, including several top reviewer awards.

Journals IEEE Transactions on Signal Processing [2026]
IEEE Signal Processing Magazine [2026]
IEEE Transactions on Aerospace and Electronic Systems [2026]
Transactions on Machine Learning Research (TMLR) [2024, 2025, 2026]
Journal of Open Source Software (JOSS) [2024, 2025, 2026]
EURASIP Journal on Advances in Signal Processing [2023]

Conf. NeurIPS [2025; **Top Reviewer** 2025]
ICLR [2025, 2026]
ICML [2025, **Position Track Gold Reviewer** 2026]
European Signal Processing Conference (EUSIPCO) [2024, 2025, 2026]
NeurIPS Workshops [2024]
IEEE Statistical Signal Processing Workshop [2025]
Asilomar Conferences on Signals, Systems, and Computers (ASSC) [2025, 2026]
AAAI [2026]

University Service

2021–25 Stony Brook University Graduate Student Organization
Primary Senator, Dept. of Electrical & Computer Engineering
Graduate Student Representative for Graduate Council
Reviewer for Distinguished Travel Award

2022–23 SBU Strategic Planning Committee [link]
Member of Strategic Planning Committee; helped draft vision statement for Project REACH

Expository Talks

2024 SBU ECE Honors Seminar
Ensembling and Fusion: Perspectives and Solutions from Signal Processing

2022 Stony Brook Math Club
Making Markov Chains with Metropolis [video]

Organizing

2026 Special Session on "Bayesian Learning and Inference on Graphs"
Co-organized with Kurt Butler, Victor Elvira, and Petar M. Djurić
Asilomar 2026, Pacific Grove, CA

2026 Minisymposium on "Sequential Inference for Dynamical Systems"
Co-organized with Fernando Llorente-Fernandez
SIAM UQ 2026, Minneapolis, MN.

2025 Special Session on "Advances in Causal Inference: Theory and Applications"
Co-organized with Petar M. Djurić
Asilomar 2025, Pacific Grove, CA

Mentoring

2026 Basis Research Institute
Luke Snow (PhD Research Intern; Wiener Space Cubature for Filtering & Seqeuntial Likelihood-Informed Subspaces)

2022–25 CUNY Directed Reading Program
Mike Prebil (*Gaussian Processes and Bayesian Optimization*)
Edgar Cuapio Diaz (*Gaussian Processes and Bayesian Optimization*)
Jonathan Jaimangal (*Hamiltonian Monte Carlo*, **Winner Outstanding Poster Award**)
Masroor Khonkhodzhaev (*Dynamic Programming and Reinforcement Learning*)
Percy Martinez (*Fourier Analysis*)
Isabella Chittumuri (*Elements of Statistical Learning*)

2023 Stony Brook DRP
Shailen Smith (*Probabilistic Machine Learning*)

2022 Bayesian Data Analysis for the Global South
Volunteer TA for online course by Aki Vehtari [certificate]

Updated October 2026